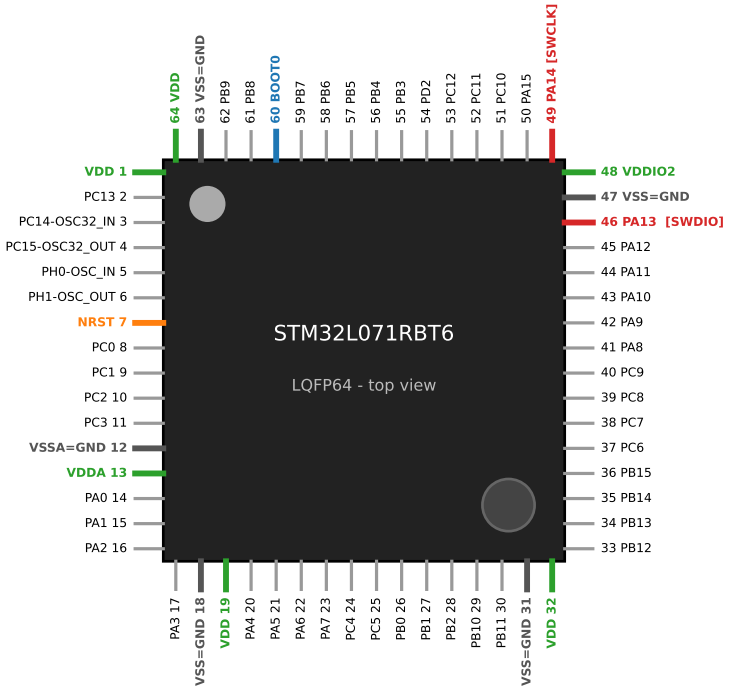
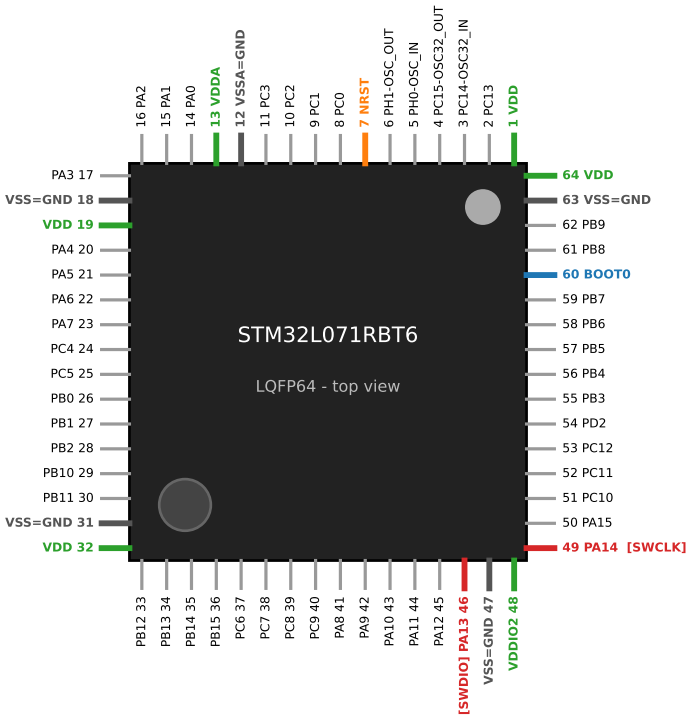
ANDRTF3 controller - STM32L071RBT6 (LQFP64) pinout

Datasheet orientation: pin-1 dimple top-left, pins count counter-clockwise



Rotated 90 deg CW: dimple top-right, as in photos/back-mcu-swd-header.jpg



SWDIO / SWCLK NRST BOOT0 VDD / VDDA / VDDIO2 GND = VSS / VSSA

Small dimple = pin 1. The big round mark is the mould ejector, NOT pin 1. The marking text on this chip is printed rotated - ignore it for orientation.

SWD connections (measured 2026-10-24, see docs/connectors.md)

Signal	MCU pin(s)	2x4 header contact	2x5 pad	STDC14 (STLINK-V3MINIE)
SWDIO (PA13)	46	row 2 right		4 T_SWDIO
SWCLK (PA14)	49	row 3 right		6 T_SWCLK
NRST	7	-	row 5 left	12 T_NRST
BOOT0	60			- (do not connect)
VDD	1 / 64 / 19 / 32	row 1 left		3 T_VCC (sense only)
GND (VSS)	63 / 47 / 18 / 31 (VS	row 4 left		5 / 7 GND
SWO: PB3 is RS485 DE	55			8 T_SWO (leave open)
TX (debug, PC12)	53	row 3 left		13 T_VCP_RX (optional)

STDC14 connector on STLINK-V3MINIE (1.27 mm, 2x7)

Pin	Signal	Pin	Signal
1	reserved (NC)	2	reserved (NC)
3	T_VCC (target voltage sense)	4	T_SWDIO / TMS
5	GND	6	T_SWCLK / TCK
7	GND	8	T_SWO / TDO
9	T_JRCLK (NC for SWD)	10	T_JTDI (NC for SWD)
11	GNDDetect	12	T_NRST
13	T_VCP_RX (probe input)	14	T_VCP_TX (probe output)

STDC14 pinout per ST STLINK-V3 user manuals (UM2448/UM2910) - the data brief on disk has no pinout; confirm pin 1 marking on the cable.

Rules

1. Continuity test with the board UNPOWERED.
2. For SWD, power the sensor from its own supply (24 V terminals). T_VCC only senses the target voltage - the STLINK-V3MINIE does not power the board.
3. Minimum wiring: SWDIO, SWCLK, GND, T_VCC. Add NRST for connect-under-reset.
4. Leave BOOT0 unconnected. It is normally pulled to GND by a resistor - note its value.
5. Nothing gets written to the chip before a verified dump exists (scripts/probe.sh, scripts/dump.sh).

Board parts for reference: ISO3082 (isolated RS485), MCP3551/3553 (PT1000 ADC, SPI), 2x 74HC165 (DIP switches S1/S2), LCD via zebra strip, B8S bridge + buck to 3.3 V.